Ex:No:2

Ex:2 - Study of Network cables.

Aim:

To study of diffrent types of Network cables and make a cross-wired cable and straight through cable using clamping/crimping tool.

a) Understand diffrent types of Network cable:

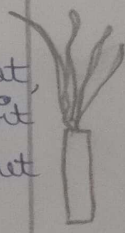
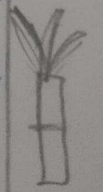
Diffrent type of cables used in networking are

i) Unshielded twisted pair (UTP) cable,

ii) Shielded twisted Pair (STP) cable,

iii) Coaxial cable,

iv) Fibre optic cable.

Cable type	Category	Maximum data transmission	Advantages/ Disadvantages:	Use	Image
UTP	Category 3	10 bps	Advantages: * cheaper in cost. * Easy to install Disadvantages: * More prone to (EMI).	10 base-T ethernet	
	Category 5	Up to 100 mbps		fast ethernet, gigabit ethernet	
	category 5e	1Gbps			
STP	Category 6,6a	10 Gbps	Advantages: * Shielded. * Faster than UTP. * Less susceptible to noise and interference.	gigabit, ethernet, 10G ethernet (55m) widely used in data centres.	

SSTP	Category 7	10 Gbps	Disadvantages: * Expensive * Greater installation effort.	gigabit ethernet, 10G ethernet.
Coaxial cable	RG-6. RG-59 RG-11	10-100Mbps.	* High bandwidth * Low loss Bandwidth Disadvantages: * Limited distance. * Cost. * Size is bulky.	Speed of signal is 500m television network high speed internet Connections.
fibre optics cable.	Single mode Multi mode	100 Gbps	Advantages: * High speed * High Bandwidth * High Security. Disadvantages: * Expensive. * Requires skilled installers.	* Maximum distance of fibre optics cable is around 100 meters.

b) Make your own ethernet Cross - Over cable / Straight cable.

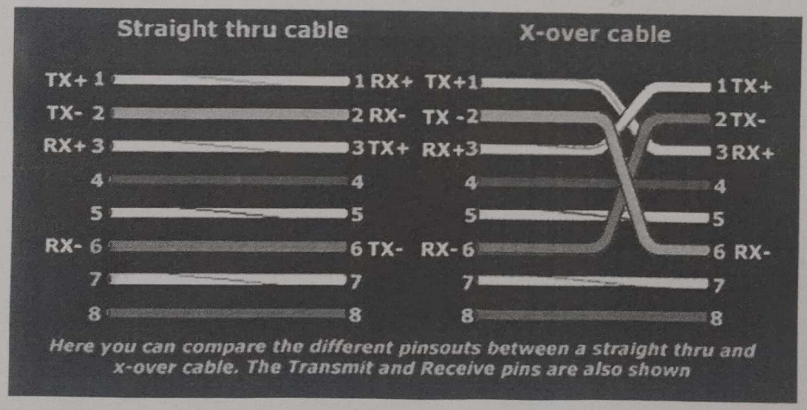
Tools and parts needed;

* ethernet cabling. CAT 5e is certified for gigabit support, but CAT 5 cabling works as well just over shorter distances.

* A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and crimp and cut the shielding off the cables.

* Two RJ45 plugs,

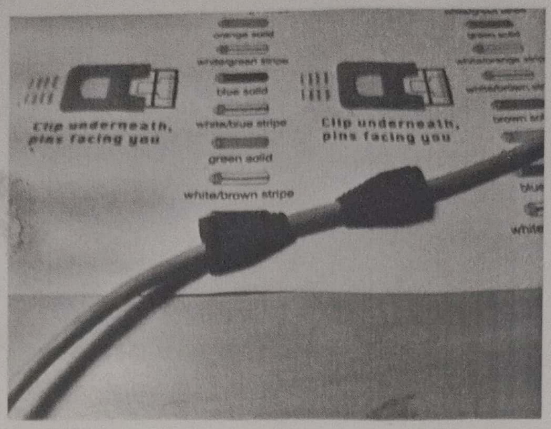
* Optional two plug shields.



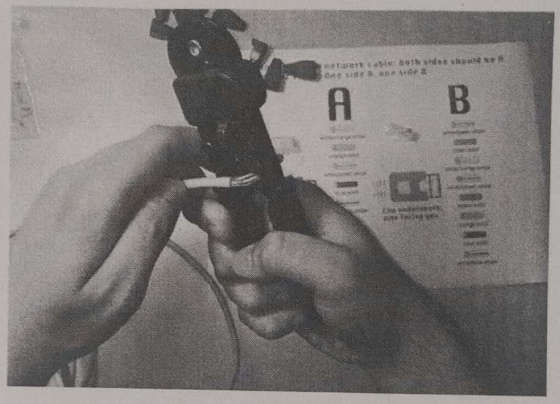
Here you can compare the different pinsouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Scanned with OKEN Scanner

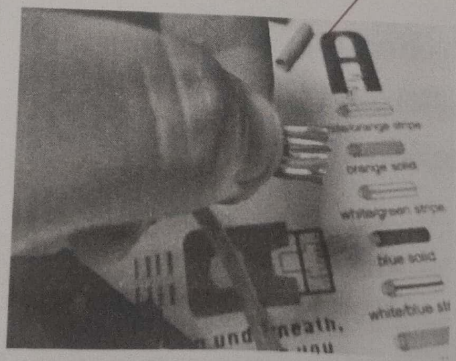
Step 1: To start construction of the device, begin by threading shields onto the cable,



Step 2: Next, strip approximately 1.5 of cable shielding from both ends. The crimping tool has a round area to complete this task.

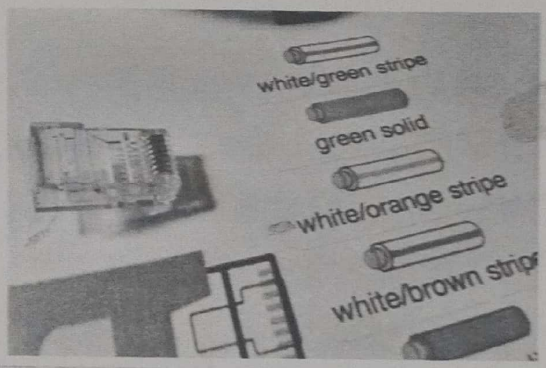


Step 3: After, you will need to untangle the wires; there should be four "twisted pairs", referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

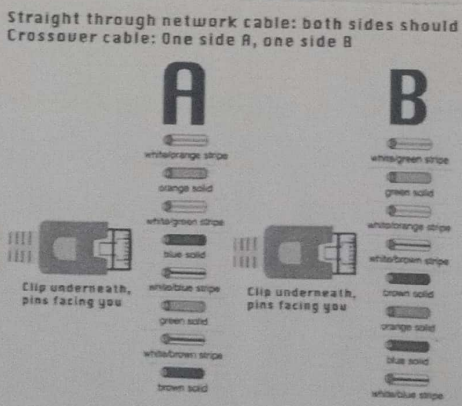


Scanned with OKEN Scanner

Step 4:
Once the order is correct, bunch them together in a line and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown



Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped off too much shielding. Simply snip the cables back a little more.



Scanned with OKEN Scanner

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B using diagram A.

To test it, plug it in and attempt to connect two devices directly.

Student observation:

1) What is the difference between Cross cable and straight cable?

⇒

	Straight cable	Cross cable.
*	Same wiring standard on both ends.	* Different wiring standards on each end.
*	Used to connect different types of devices.	* Used to connect similar types of devices.
*	Eg: PC to switch, Router to Switch.	* Eg: PC to PC, switch to switch.

2) which type of cable is used to connect two PCs?

⇒ Cross cable because both PCs are similar devices, so the transmit (TX) and receive (RX) wires must be crossed to communicate directly.

Scanned with OKEN Scanner

3) which type of cable is used to connect a router/ switch to your PC?

⇒ Straight cable, because a PC and a switch/ router are different types of networking devices.

4) Find out the category of twisted pair cable used in your lab to connect the PC to the network socket.

⇒ Category: Cat5e or Cat6.

5) write down your understanding, challenges faced, and output received while making a twisted pair cross / straight cable.

⇒

* I learned the difference between straight and cross cables.

* I understood the T568A and T568B wiring standards.

* Arranging the wire colors in the correct order.

* Successfully prepared a straight / cross twisted pair cable.

* Tested the cable using a cable tester.